

VLS 804: Buck Converter Compensation

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Please download the `Lec18_Buck.asc` and `Lec18_AC_Buck.asc`. It is an unstable 2.5-V to 1.8-V asynchronous buck converter with a schottky diode. First spend some time analyzing the loop and the model.

- (a) What is the output filter transfer function, $H_{RLC}(s)$? Refer to Lecture 7 for part (a), (b) and (c).
- (b) What is the modulator gain? Assume the maximum duty cycle to be 100%.
- (c) What is the duty-cycle to output transfer function $G_{vd}(s)$?
- (d) As you would have guessed, we want you to make this converter stable and achieve a phase margin of over 45° and bandwidth higher than 10 kHz. You cannot change the power path parameters. You have the liberty to change the compensation parameters. First simulate the `Lec18_AC_Buck.asc` and stabilize the linearized model. Submit the gain and phase plot of the stable converter. Also, save as a new schematic with the new compensator component values.
- (e) Now, use these values in the `Lec18_Buck.asc` file and run a transient simulation when the supply starts up (`.tran 1m startup`) on the supply. Does the converter become stable? Check and save the output voltage response.